Data Structures and Algorithms Assignment (JavaScript)

1. Linear Data Structures (JavaScript Code)

Array:

```
let arr = [10, 20, 30, 40, 50];
arr.forEach(e => console.log(e));
console.log(arr[2]);
```

Stack:

```
let stack = [];
stack.push(10);
stack.push(20);
console.log(stack.pop());
console.log(stack[stack.length - 1]);
```

Queue:

```
let queue = [];
queue.push(10);
queue.push(20);
console.log(queue.shift());
```

Linked List:

```
class Node {
  constructor(data) {
    this.data = data;
    this.next = null;
  }
}

let head = new Node(10);
head.next = new Node(20);
head.next.next = new Node(30);

let temp = head;
while (temp) {
  console.log(temp.data);
  temp = temp.next;
}
```

2. Non-Linear Data Structures

Binary Tree:

```
class TreeNode {
  constructor(value) {
    this.value = value;
    this.left = null;
    this.right = null;
  }
}

let root = new TreeNode(10);
root.left = new TreeNode(5);
root.right = new TreeNode(15);

function inorder(node) {
```

```
  if (node) {
    inorder(node.left);
    console.log(node.value);
    inorder(node.right);
  }
}
inorder(root);
```

Graph:

```
let graph = {
  A: ['B', 'C'],
  B: ['A', 'D'],
  C: ['A'],
  D: ['B']
};

function dfs(node, visited = new Set()) {
  if (!visited.has(node)) {
    console.log(node);
    visited.add(node);
    graph[node].forEach(n => dfs(n, visited));
  }
}
dfs('A');
```

3. Applications of Data Structures

Arrays: Used in databases for fast access. Linked Lists: Used in memory management. Stacks: Used in undo operations. Queues: Used in scheduling systems. Trees: Used in file systems. Graphs: Used in social networks and maps.

4. Real Applications

Web browsers use stacks for navigation. Social media uses graphs for connections. Navigation systems use graphs for shortest path. Operating systems use queues for scheduling. Text editors use stacks for undo/redo.

5. How Data Structures and Algorithms Work

Data structures store data while algorithms process it. Efficient systems depend on choosing the right structure and algorithm. Time complexity measures speed. Space complexity measures memory usage.